

26-Week Work Plan — Revision D

Three-phase AMI meter • STM32U575VGT6 + ADE9000ACPZ-RL • in-house KiCad design, two spins, **no evaluation board — spin 1 is our eval board**
Team: 5 engineers (onboarding week complete) + program lead + two part-time mentors (firmware, hardware/analog) • Horizon: weeks 1–26 • Supersedes Revision C

What changed in this revision

1. No EVAL-ADE9000 board will be purchased; the team goes straight to its own hardware. **2.** The one-week embedded onboarding is already complete, so week 1 starts on real work rather than training. Together these pull the spin 1 fabrication release forward by roughly three weeks and add real margin at the back end — but they also mean no metrology firmware can run on hardware until spin 1 arrives.

1. The Two Changes, and What They Cost

Change	Consequence	How this plan absorbs it
Onboarding week already complete	The team can read a schematic symbol, drive a GPIO, and use a debugger. Weeks 1–2 of Revision C are freed	Schematic capture starts in week 2 instead of week 5. Every downstream milestone moves earlier, which is where the new back-end margin comes from
No ADE9000 evaluation board	There is no hardware on which to develop or validate metrology firmware until spin 1 boards arrive. The ADE9000 register map, SPI transactions and calibration logic must be written blind	Two mitigations, both in section 3: the driver is written against the datasheet with a host-side mock and a test suite, and spin 1 is built in quantity so it becomes the eval board for the whole team
Spin 1 is now on the critical path for firmware as well as hardware	In Revision C a fabrication slip only delayed hardware. Now it stalls the metrology lane too	Spin 1 is deliberately kept simple and releases in week 5. Section 3 keeps three firmware lanes productive during the fab window

One honest note on cost

A batch of spin 1 boards will not be cheaper than a single evaluation board — PCB, assembly, stencil and components at low volume usually exceed it. The justification for going direct is **capability and schedule**: the team learns the full design flow earlier, and the board matches our actual sensing design rather than ADI's demonstration one. If the decision was made purely on cost, it is worth re-checking the arithmetic before the fab order goes out.

2. Board Definitions

	Spin 1 — our evaluation board (low voltage)	Spin 2 — full mains board
Purpose	Replaces the ADE9000 eval board. The firmware development platform for the whole team, and the team's first complete KiCad design	The three-phase meter prototype
Contents	STM32U575VGT6, ADE9000, SPI and interrupt wiring, CT input conditioning, divider footprints, FRAM, display and debug headers, test points on every analog node, bench DC input	Everything from spin 1, plus wide-range SMPS with diode-OR, terminal block, surge protection, supercapacitor and battery backup, tamper inputs, contactor driver, full creepage and clearance
Power	Bench DC only. No mains on this board, ever	230/400 V three-phase, under the section 8 protocol

	Spin 1 — our evaluation board (low voltage)	Spin 2 — full mains board
Signal source	Isolated low-voltage AC source or direct signal injection into the CT and divider inputs	Real three-phase source through the terminal block
Quantity	8–10 units — one per engineer, plus spares for destructive or rework experiments	5–6 units
Deliberate additions	Generous test points, series resistors and 0 Ω links on every interface, socketed or accessible key nets. This board is for probing, not for looking tidy	Test points retained; production-style layout otherwise
Who leads layout	E4, paired with the hardware mentor (M2)	M2 owns the mains-side layout personally ; E4 owns digital and analog
Releases to fab	Week 5	Week 16
Boards expected	Week 8–9	Week 19–20

Quantity is the cheapest risk reduction in this plan

Two boards means the team queues for hardware and one mistake halts a lane. Eight to ten boards means every engineer develops in parallel, someone can cut a track to test a theory, and a destroyed board is an afternoon rather than a fortnight. At low volume the marginal cost per extra board is small compared with the assembly setup you are paying for anyway.

3. Firmware Before Hardware Exists

Weeks 1–8 have no ADE9000 hardware in the building. That time is not lost — it is spent building the parts of the metrology firmware that do not need a chip, in a form that can be tested on a workstation. This is the same discipline the onboarding week established with the command parser: separate logic that can be tested on a PC from code that must touch a register.

Work item	What it is	How it is verified without hardware
Register map header	Every ADE9000 register, address, bit field and reset value transcribed from the datasheet into a header file	Second-person review against the datasheet, line by line. This is tedious and it is exactly the kind of error that costs a week during bring-up
SPI transaction layer	Read, write, and burst-read functions with the device's framing and CRC handling	Unit tests on the host against a mock SPI that replays known byte sequences from the datasheet's timing examples
Device mock	A software model of the ADE9000 that answers register reads with configurable values and can inject synthetic waveform data	Lets the whole metrology pipeline — read, scale, accumulate, report — run and be tested on a workstation in CI
Scaling and accumulation logic	Conversion from raw codes to volts, amps, watts and kWh, including the fixed-point arithmetic	Host tests with known inputs and hand-computed expected outputs. Overflow and rounding behaviour tested here, not discovered on hardware
Calibration procedure draft	The sequence and the arithmetic for gain, phase and offset calibration per channel	Dry-run against the mock with simulated errors injected, so the procedure is exercised before it meets a real board
Bring-up script	The exact ordered list of what to read and write on first power-up, with expected values	Reviewed by both mentors. Turns day one with a new board from improvisation into a checklist

When spin 1 arrives in week 8, the team should be swapping the mock for the real SPI driver, not starting to write a driver. If the mock work is done properly, the first meaningful register read happens within days of the boards landing rather than weeks.

Where to get the reference material without buying the board

ADI publishes the ADE9000 evaluation board schematic, layout guidance and user guide as free documentation. The reference schematic is the starting point for our analog front end whether or not we own the hardware — download it in week 1 and design from it rather than from a blank sheet.

4. Definition of Success at Week 26

#	Exit criterion
1	Spin 1 designed in KiCad, fabricated in quantity, brought up, and serving as the team's development platform
2	Spin 2 full mains board designed, fabricated, and safely powered from a three-phase source
3	ADE9000 reading all seven channels on spin 2 from the board's own CTs and dividers
4	Three-phase energy accumulation running on own hardware, with an accuracy comparison against a reference
5	Calibration procedure executed at least once on real hardware

#	Exit criterion
6	Tamper inputs, display and local console working on the assembled unit
7	DLMS/COSEM read-only association: a test client reads billing registers and the event log
8	Power-fail sequence proven on spin 2: last-gasp write to FRAM, state intact across interruption
9	Documentation set: KiCad projects in version control, review records, ERC/DRC reports, BOMs, DFM correspondence, bring-up logs, defect registers for both spins, host-side test suite
10	Phase-2 plan: spin 3 scope, full V&V, PLN Pusertif / legal metrology / TKDN timeline and cost

5. Gates

Gate	Week	Exit criteria
G0 — Requirements and setup	1	Requirements v1.0 frozen; KiCad project and CI running; PCBA manufacturer selected with design rules in hand; ADI reference documentation obtained
G1 — Spin 1 release to fab	5	Schematic reviewed by both mentors; ERC clean; layout reviewed for mixed-signal integrity; DRC clean against manufacturer rules; BOM availability verified; DFM returned and resolved
G2 — Spin 1 alive	10	Boards powered from bench DC; MCU boots; SPI link to ADE9000 working; channels respond to injected signals; mock replaced by the real driver
G3 — Spin 2 release to fab	16	Schematic reviewed with mains-side scrutiny; creepage and clearance formally checked against IEC 62052-11 ; EMC design review passed; DRC clean; DFM resolved
G-S — Electrical safety gate	19	Mandatory before the first mains power-up. Isolation transformer and RCD installed; training signed off; written power-up procedure reviewed; two-person rule agreed; M2 present
G4 — Spin 2 alive on mains	21	Board safely powered from three phases; rails verified; all seven channels reading from the board's own sensors
G5 — Handoff	26	Soak passed; documentation set delivered; Phase-2 plan presented

6. Team Roles

ID	Role	Weeks 1–8 (no hardware)	Weeks 9–26 (hardware available)
E1	Metrology and calibration	Register map, scaling and accumulation logic, device mock, calibration procedure draft — all host-tested	Swap mock for real driver on spin 1; calibration on hardware; accuracy work on spin 2
E2	Protocol and data model	DLMS read-only association on the existing VGT6 board; OBIS model for three-phase registers	Port to spin 1, then spin 2; event log and billing registers readable end to end
E3	Application, display, tamper	Display, console and tamper event manager on the existing VGT6 board; digital schematic sheets	Port to the new boards; tamper integration with the ADE9000 detections
E4	Hardware workstream lead	KiCad project ownership, schematic integration, layout, DFM, manufacturer interface for spin 1, then spin 2	Bring-up lead for both spins; test bench; defect registers
E5	QA, documentation, coordination	BOM and sourcing, DFM correspondence, traceability, host-test CI, safety records	Bring-up log discipline; review minutes; documentation set
PL	Program lead	Manufacturer relationship, procurement, PLN engagement, safety accountability	As before, plus Phase-2 preparation
M1	Firmware mentor	Architecture and code review; reviews the register map and mock design	Bring-up support; code review
M2	Hardware / analog mentor	Schematic and layout review; pairs with E4 on spin 1 layout	Owens mains-side layout of spin 2 ; safety officer; present at G-S and first mains power-up

7. KiCad Discipline (carried forward, unchanged)

- **Version control from day one.** KiCad project in Git alongside the firmware; every review and fab release is a tagged commit.
- **ERC and DRC in CI.** `kicad-cli` runs on every push, exactly as the firmware tests do; a failing check blocks the merge.
- **Library discipline.** Custom symbols and footprints for the ADE9000 (LFCSP) and STM32U575VGT6 (LQFP100) built once, verified against datasheet drawings by a second person.
- **Footprint verification ritual.** Every new footprint printed 1:1 and physically checked against the real component before layout. This single habit prevents the most common and most expensive novice error — and with no eval board to fall back on, a wrong footprint now costs the whole metrology lane, not just a board.
- **Manufacturer rules first.** Stackup, trace and space, drill sizes, test point and fiducial requirements encoded as KiCad design rules before layout begins.
- **Two-person review of every sheet.** No schematic sheet goes to layout reviewed only by its author.
- **Four-layer stackup minimum**, with a continuous ground reference under the analog section and under the SPI bus.
- **DFM round trip before the fab order**, with a week budgeted on each spin.

8. Electrical Safety Protocol

Unchanged, and it overrides schedule pressure in every case

Spin 1 carries no mains, which is why it is first. Spin 2 does. From week 19 onward, five engineers with no electrical experience are working on a 400 V board they designed themselves. Rules: formal training signed off before bench access • isolation transformer and RCD on every mains bench • two-person rule, no exceptions • written power-up procedure signed for each new revision • prove dead before touching • enclosed fixture for soak tests • M2 is safety officer with E4 deputising, either may halt work without justification.

9. Week-by-Week Plan

Phase 1 — Spin 1 design and blind firmware (weeks 1–5)

Wk	E1 metrology (blind)	E2 / E3 firmware	E4 hardware lead	E5 QA / procurement
1	ADE9000 datasheet study; register map header begun; Gate G0	DLMS and display work continues on the existing VGT6 board	KiCad project and CI set up; ADI reference schematic obtained; custom symbols and footprints begun	PCBA manufacturer selected; design rules obtained; long-lead parts ordered (ADE9000, CTs, FRAM)
2	Register map complete and second-person reviewed; SPI transaction layer	OBIS model; tamper event manager	Spin 1 schematic capture: analog front end adapted from the reference, with E1 on the analog sheets	BOM v1 with availability check; footprints verified 1:1 against real parts
3	Device mock; scaling and accumulation logic with host tests	Display screens for three-phase quantities	Schematic complete; full review with both mentors; corrections applied; ERC clean	Review minutes; action tracking; test point list agreed
4	Calibration procedure drafted; dry-run against the mock	Console command set for metering data	Layout , paired with M2; DRC clean; DFM submitted	DFM findings tracked to closure; assembly documentation
5	Bring-up script written and reviewed by both mentors	Host-side test suite green in CI	DFM resolved. Gate G1 — spin 1 released to fab, 8–10 units	Fab package archived and tagged; incoming inspection checklist

Phase 2 — Fabrication window and spin 1 bring-up (weeks 6–10)

Wk	Activities	Notes
6–7	Spin 1 in fabrication. Metrology pipeline fully exercised against the mock in CI. DLMS, display, tamper and console matured on the existing VGT6 board. Test bench assembled: bench supplies, isolated low-voltage AC source, signal injection rig, reference meter	Firmware lanes are productive because of section 3. Without the mock work, this fortnight would be idle
8	Spin 1 boards received. Incoming inspection; bench-DC power-up of one unit following the bring-up script; MCU boot and SWD access confirmed	One board first, not all of them. Find the fault on unit one
9	SPI link to ADE9000 established; mock replaced by the real driver ; first real register reads; remaining units built up and distributed	This is the moment the blind work pays off, or does not
10	Signal injection into the CT and divider inputs; per-channel RMS and power verified; defects logged for spin 2. Gate G2 — spin 1 alive	Every engineer now has their own board

Phase 3 — Spin 2 design (weeks 11–16)

Wk	Firmware on spin 1 (E1–E3)	Spin 2 design (E4 + M2)	Program / QA
11–12	Energy accumulation across three phases; calibration method exercised on real silicon; display and console ported	Spin 2 schematic: SMPS with diode-OR, surge protection, terminal block, backup, contactor driver — M2 leads the mains sheets	Spin 1 defect register closed out into the spin 2 change list
13–14	Tamper detections integrated with ADE9000 reverse-energy and neutral-loss flags; DLMS read set frozen	Schematic complete; creepage and clearance check against IEC 62052-11 ; EMC design review	Safety gate preparation begins: transformer and RCD installed, procedure drafted

Wk	Firmware on spin 1 (E1–E3)	Spin 2 design (E4 + M2)	Program / QA
15–16	Firmware consolidation; host test coverage extended; documentation catch-up	Spin 2 layout — M2 owns the mains side, E4 the digital and analog. DRC clean; DFM resolved. Gate G3 — release to fab	Three-phase source booked for weeks 21–24; fab package tagged

Phase 4 — Mains bring-up, integration and handoff (weeks 17–26)

Wk	Activities
17–18	Spin 2 in fabrication. Continued firmware maturation on spin 1; safety fixture built; power-up procedure finalised and reviewed
19	Spin 2 boards received; inspection; low-voltage digital-only power-up first. Gate G-S — electrical safety gate, M2 present
20	First mains power-up under written procedure; rail checks; SMPS and diode-OR verification; SPI integrity on the new board
21	All seven channels reading from the board's own CTs and dividers; phase sequence and neutral verified. Gate G4
22	Calibration executed on hardware: gain, phase and offset per channel, with the metrology consultant present
23	Accuracy comparison against the three-phase reference across load and power factor; results recorded
24	Tamper, display, DLMS read and the power-fail sequence verified end to end on spin 2
25	Soak test in the safety enclosure; defect triage; feature freeze; documentation sprint
26	Phase-2 plan presented; final demo and handoff. Gate G5. Retrospective

10. Risk Table

Risk	Likelihood	Mitigation / pre-agreed response
Injury during mains working (week 19 onward)	Low but unacceptable	Section 8 in full. G-S is a hard stop. M2 or E4 may halt work at any time without justification. Never traded against schedule
Spin 1 has a fault that blocks the metrology lane	Medium — new in this revision	With no eval board there is no fallback platform. Mitigations: generous test points and 0 Ω links for isolation and rework; 8–10 units so a damaged board is not fatal; bring-up script reviewed in advance; M2 available in week 8
Register map transcription error found during bring-up	Medium	Second-person review in week 2 against the datasheet. If found late, it is a firmware fix rather than a board fix — which is why the mock work is worth doing even though it is tedious
Footprint or symbol error reaches fabrication	Medium	1:1 print-and-check ritual. Now more costly than in Revision C: a wrong ADE9000 footprint stalls both hardware and firmware
Spin 2 needs a third spin	Medium–high	Accepted. Spin 3 is a Phase-2 item. Blue-wire fixes and a documented change list are the Phase-1 deliverable
Layout quality insufficient for class 0.5S	Medium	Analog section adapted from the ADI reference layout. If measured performance falls short, re-baseline to class 1.0 at G4 rather than chasing it into Phase 2
Fabrication slips on either spin	Medium	Weeks 17–18 are buffer. Beyond two weeks' slip, reduce the week 23 accuracy comparison to a spot check at a few load points and protect bring-up and calibration
M2 unavailable	Medium	Program-stopping for the mains phase: G-S cannot pass without the safety officer, and the spin 2 mains layout is M2's personal deliverable
Scope pressure to re-add STS or full DLMS	Medium	Baseline is section 4. Additions only via a gate review stating explicitly what is displaced

11. Program Lead Checklist

- **Week 1:** PCBA manufacturer selected and design rules obtained in writing; both mentors' schedules confirmed; ADI reference documentation downloaded; long-lead parts ordered (ADE9000, CTs with phase-error data, FRAM, reference meter)
- **Week 1:** re-check the spin 1 batch cost against what an evaluation board would have cost — if the gap is large, the decision deserves one more look before the fab order
- **Week 3:** attend the spin 1 schematic review personally
- **Week 5 and week 16:** confirm DFM findings are resolved *before* each fab order is placed
- **Week 8:** ensure M2 is present for spin 1 first power-up — the first board is where the expensive mistakes surface
- **Week 14:** attend the spin 2 creepage and clearance review; this is a safety review, not a design review
- **Week 19:** chair the safety gate G-S. Do not delegate it
- **Week 21:** Phase-2 budget conversation opened with management
- **Week 26:** handoff review and Phase-2 decision

12. What This Revision Asks You to Accept

Revision D is the tightest-coupled plan of the four. Removing the evaluation board removes the safety net underneath the metrology lane: if spin 1 has a serious fault, firmware and hardware stall together. In exchange, the completed onboarding buys back three weeks, so the back end of the schedule now has genuine margin — calibration and the accuracy comparison sit in weeks 22–23 rather than being squeezed against the handoff.

Two things decide whether this works. The first is the blind firmware work in section 3: done properly, spin 1 goes from boards-in-hand to real register reads in days. Done carelessly, the team starts writing a driver in week 9 and the plan loses its margin immediately. The second is building spin 1 in quantity — it is the cheapest insurance available against the risk this revision introduces.

Prepared by Edy Gunawan, Program Lead. Revision D supersedes Revision C. Supporting detail in the three-phase meter preliminary design document.